

Gavin Abrigo

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MLOps, Data Engineering, AI Engineering, Project Management, Agentic Workflows

EDUCATION

University of California Merced

Bachelor of Science in Computer Science & Engineering

Merced, CA

May 2027

TECHNICAL SKILLS

Programming Languages — TypeScript, JavaScript, GoLang, Java, Python, C/C++, HTML/CSS, LaTeX, MIPS Assembly, SQL, MATLAB, SPARQL

Data & Databases: — PostgreSQL, SQLite, MongoDB, Excel, Databricks

Frameworks & Tools: — Shell, React.js, FastAPI, Flask, REST APIs, Git, GitHub, Node.js, Next.js, Vite, StreamLit, Docker, CI/CD, JSON

AI & ML Tools — Nvidia NIM, Tesseract, OpenCV, PyTorch, Pandas, LLM API, Claude API, OpenAI API, Prompt Design

EXPERIENCE

Undergraduate Researcher Intern

September 2025 - Present

LLM Knowledge Graph Retrofitter (KGR) - Prof. Pandey

Merced, CA

- Improve factual accuracy to reduce hallucinations in NLP models. Building a Python-based Knowledge Graph Retrofitting (KGR) and Retrieval-Augmented Generation (RAG) pipeline using Wikidata and GraphRAG.
- Fine-tuning entity linking systems and evaluating LLM verification performance with NLP metrics such as exact match and F1 on benchmark datasets.

Software Engineer Intern

January 2026 - May 2026

BART - Software Engineering Capstone

Merced, CA

- Automatically classify infrastructure incident images submitted via web/mobile app, utilizing structured metadata, and route to appropriate maintenance teams
- Developing image embedding and similarity detection workflows (Google Gemini) to de-duplicate submissions with postgres vector embeddings, extract contextual location data, and reduce manual review overhead.

Software Engineer Intern (backend)

January 2025 - March 2025

OculusIO (NFT Automation Software Startup)

Remote

- Developed Bash and TypeScript CLI automation tools to streamline multi-language builds, integrating Go protobuf generation and client compilation—cutting setup time by 65% and build errors by 40%.
- Implemented cross-platform environment validation scripts for Windows, macOS, and Linux, standardizing workflows and improving developer onboarding efficiency across a 6-person team.

PROJECTS

SMART BART | Software Engineering Capstone | Python, Next.js, FASTAPI, AI fullstack | May 2026

- Image-based incident reporting system for Bay Area Rapid Transit (BART) that auto-classifies submitted maintenance photos and routes to correct department
- Gives maintenance workers a unified workflow for turning scattered photo reports into organized maintenance intelligence.
- Used vector embeddings, SentenceTransformers, and cosine similarity to cluster duplicate incident photos, so BART staff see one grouped report

NYC Taxi Trip Duration Regression and Digit Classification | Python, Scikit-learn, Pandas | Oct 2025 - Dec 2025

- Predicted total ride durations for NYC taxi trips, classified credit card fraud, and multi-digit MNIST images using Python, Pandas, and scikit-learn pipelines
- Building and evaluating classification and regression models across MNIST, Taxi Trip Duration,
- Implemented logistic regression with hyperparameter tuning and reusable data-preprocessing module; built XGBoost model with 5-fold stratified cross-validation, feature visualization, and outlier-cleaning using Python and Pandas